Mahmoud Sameh

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github.com/mah-sam | linkedin.com/in/mah-sam | kaggle.com/mah01sam/code

- AI Researcher and Electrical Engineer, graduated **top of class** with a **4.95/5 GPA** from Islamic University of Madinah; **IELTS Band 8.0**.
- Research in foundation model adaptation, computer vision, and applied ML, with **three published papers** and **one under review**.
- Selected among the **top 100 nationally** for the KAUST AI Summer Training Program.
- Co-winner of **SDAIA's Enjaz Hackathon** (150K SAR prize).
- Awarded **2nd Best Graduation Project** in AI university-wide.

Education

Bachelor of Science (B.S.) in Electrical Engineering — August 2020 – June 2025
Islamic University of Madinah, Madinah, Saudi Arabia

- GPA 4.95/5 (Academic Record) (External Courses)
- Accredited Engineer, Saudi Council of Engineers (SCE) – Membership No. 1213941

Language Proficiency

IELTS Academic: Overall Band **8.0** (L: 8.5, R: 8.5, W: 7.0, S: 7.0) (Certificate)
Duolingo (DET): Score **150**/160 (Certificate)

Work Experience

AI Engineer (Full-time Internship) — October 2025 – February 2026
Aajil Fintech

- Built AI infrastructure to automate credit risk assessment, reducing manual review overhead.
- Developed a feature-engineering pipeline that generates financial analyses from raw application data.
- Trained and deployed classification/extraction models with end-to-end MLOps.
- Bridged SMEs and the AI team, translating domain feedback into actionable model improvements.

AI Co-Innovation Program — September 2025
Aramco & RDIA

- Participated in a month-long applied AI program focused on **methane and gas detection** from remote sensors.
- Investigated the use of **hyperspectral imaging** and deep learning for spectral-signature identification of fugitive emissions, building on prior HSI hardware/software experience at KFUPM.

Research Intern (Full-time Internship) — June 2025 – August 2025
King Fahd University of Petroleum and Minerals

- Designed a complete experimental protocol for a novel hyperspectral imaging dataset of date fruits, covering sample scanning, oven-drying ground truth, and data management.
- Developed the full software stack for the lab's push-broom HSI hardware, integrating camera control, stepper motor synchronization, and data acquisition into a single Python application.
- Created an open-source GUI (PyQt6) for hyperspectral data calibration, cube assembly, and analysis, published as a first-author paper in *SoftwareX* (doi: 10.1016/j.softx.2026.102680).

Engineering College Student Contributor (Part-time) April 2023 – June 2025
Islamic University of Madinah

- Managed the Engineering College's social media, producing content that increased student engagement.
- Led student teams on college-level technical projects, offering mentorship and coordination.
- Provided student-representative feedback to faculty on program initiatives and curriculum.

Publications

[1] A. M. Alenezi, **M. Sameh**, M. Aljohani, and H. S. Alharbi, "Predicting emergency department patient arrivals at hospitals using machine learning techniques," *MDPI Healthcare*, 2026, doi: 10.3390/healthcare14091191.

[2] A. BenAbdennour, **M. Sameh**, B. A. Khawaja, A. K. Vallappil, A. M. Alenezi, Q. H. Abbasi, and S. Qazi, "Real-time human fall detection from video using YOLOv11 with pose estimation: A paradigm shift towards efficient transformer-based architectures," *IEEE Access*, 2026, doi: 10.1109/ACCESS.2026.3674843.

[3] **M. Sameh**, A. BenAbdennour, and J. K. Ali, "Efficiently adapting SAM 2 for automated schematic capture from hand-drawn circuit diagrams," *IEEE Trans. on CAD of Integrated Circuits and Systems*, no. TCAD-2025-0847, revision under preparation. [Abstract]

[4] **M. Sameh**, A. Albeladi, and A. Fawzy, "HSI Control Suite: An integrated GUI for operating and acquiring data from DIY push-broom hyperspectral imaging systems," *SoftwareX*, 2026, doi: 10.1016/j.softx.2026.102680.

Skills

AI/ML: Deep Learning, Computer Vision, Document Intelligence, NLP/LLM Integration, Data Preprocessing
Languages: Python (PyTorch, TensorFlow, Pandas, OpenCV), C, SQL, JavaScript
Engineering: Control Theory, Digital Signal Processing, Embedded Systems, Robotics
Tools: MATLAB/Simulink/Simscape, Git, Django, HTML/CSS
Project Portfolio: LinkedIn Projects | GitHub Repositories | Website

Courses

- KFUPM: Research Skills (10-Hour Course) — June 2025
- KAUST Academy: Intro to AI & Advanced AI (Passed All 3 Stages) — February 2025
- Essentials of Survey Methodology in Scientific Research — April 2024
- Mathematics for ML & Data Science Specialization (2 Courses) — October 2023
- Deep Learning Specialization (5 Courses, deeplearning.ai) — August 2022 – August 2023
- Harvard CS50: AI with Python, Intro to CS, Understanding Technology — July 2021 – August 2022
- Michigan Python Specialization (5 Courses) — December 2020 – May 2021
- Entrepreneurship Program (36-Hour Intensive) — April 2021

Full list with certificates: Direct Link

Participation & Acknowledgment

- Aramco & RDIA: AI Co-Innovation (Certificate) — September 2025
- Acknowledgment of Academic Excellence (Certificate) — May 2025
- Letter of Appreciation, Control Lab Redesign (Letter) — April 2025
- 13th International Cultural Festival (Certificate) — April 2025
- KSU Engineering Day Competitions (Certificate) — February 2025
- Enjaz Hackathon (Certificate 1) (Certificate 2) — January 2025
- WRO Saudi 2023 RoboSport (Certificate) — October 2023
- International Conference on Cyber Terrorism (Certificate) — December 2022